

## PHY102 Electricity and Magnetism Jan-April 2026: Assignment 10

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To be discussed in the Tutorial on April 24, 2026

1. Consider a square in the  $x - y$  plane with its corners at  $(\pm a/2, \pm a/2)$ . A charge  $+q$  is placed at  $(a/2, a/2)$  and  $(-a/2, -a/2)$  and a charge  $-q$  is placed at the other two corners. What are the monopole, dipole and quadrupole moments of this charge distribution.
2. Using the law of transformation of electric and magnetic fields, show that  $\vec{E} \cdot \vec{B}$  is independent of the frame of reference.
3. Consider a sphere with uniform polarization (dipole moment per unit volume). Take the polarization to be along  $\hat{z}$ . What kind of surface charge distribution will this lead to? What is the overall characteristic of this sphere in terms of moments? What kind of electric field is generated?
4. At the North pole, the earth's magnetic field is vertical and has a strength of 0.62 Gauss. The earth's field at the surface and outside earth is approximately that of a dipole. Imagine that this field is being caused by a current ring in the equatorial plane with radius at 3000km (roughly half the radius of the earth). Calculate the current that will generate such a field. What is the dipole moment associated with this system?